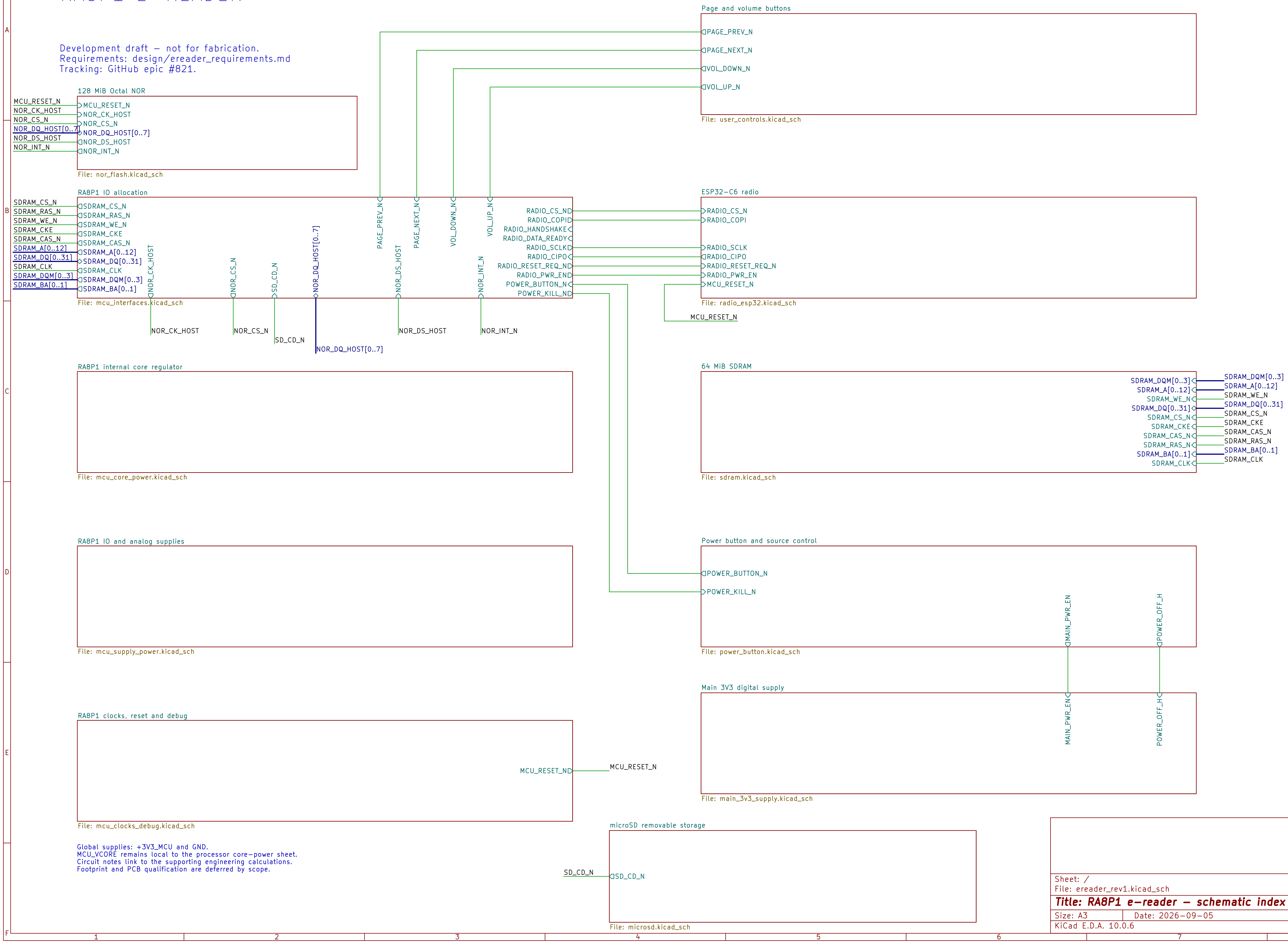
RA8P1 E-READER

Development draft — not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.



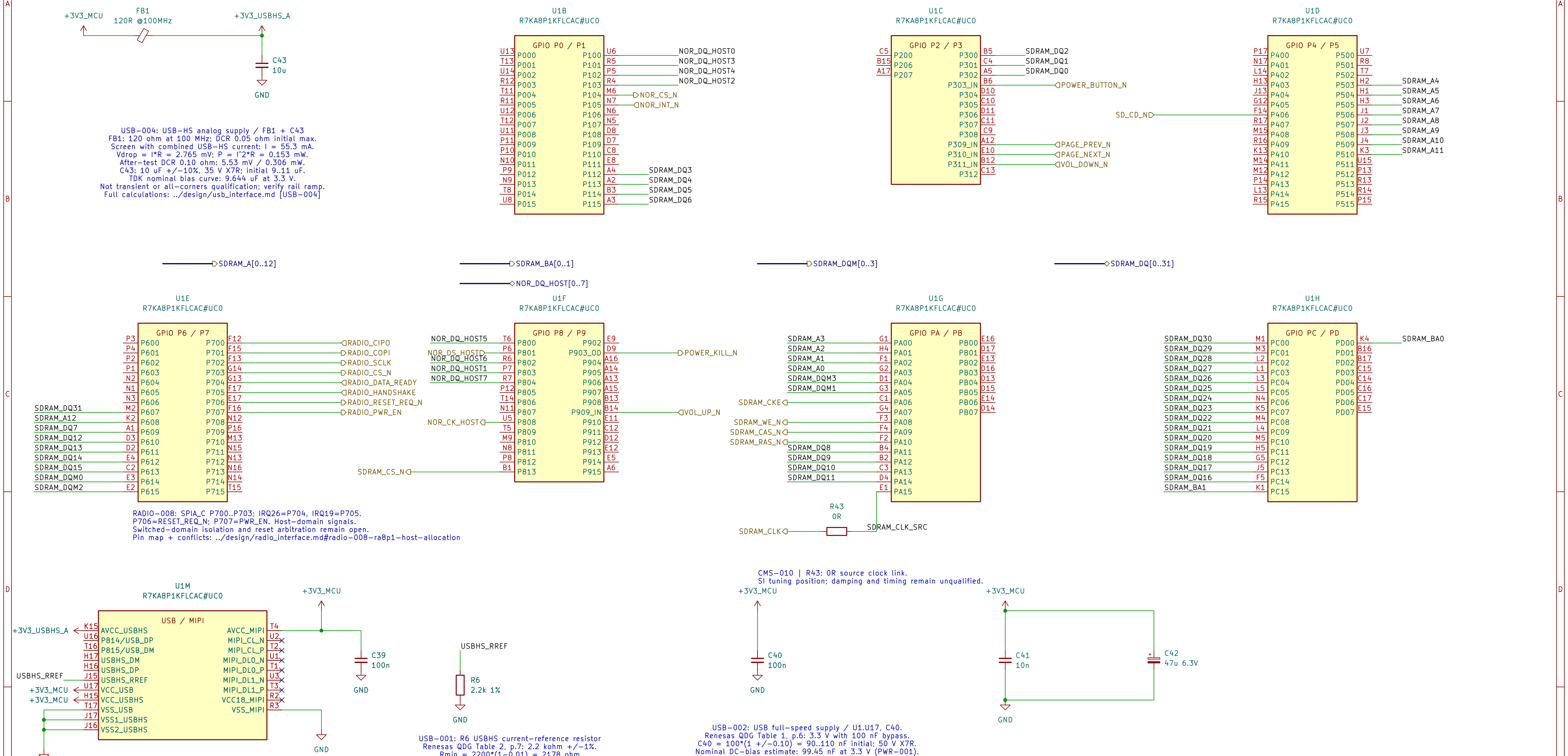
Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

Sheet: /
File: ereader_rev1.kicad_sch

Title: RA8P1 e-reader — schematic index

Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 1/12

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm: 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9..11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]

RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation

CMS-010 | R43: 0R source clock link.
SI tuning position; damping and timing remain unqualified.

MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.

USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md

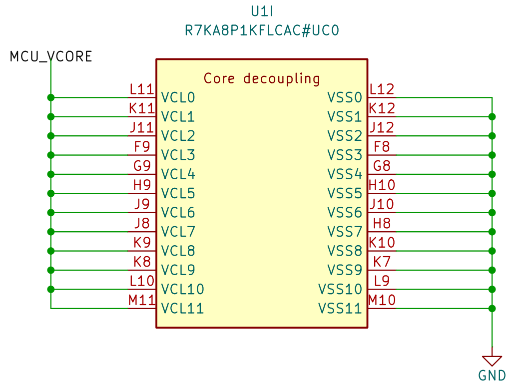
USB-002: USB full-speed supply / U1.U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md

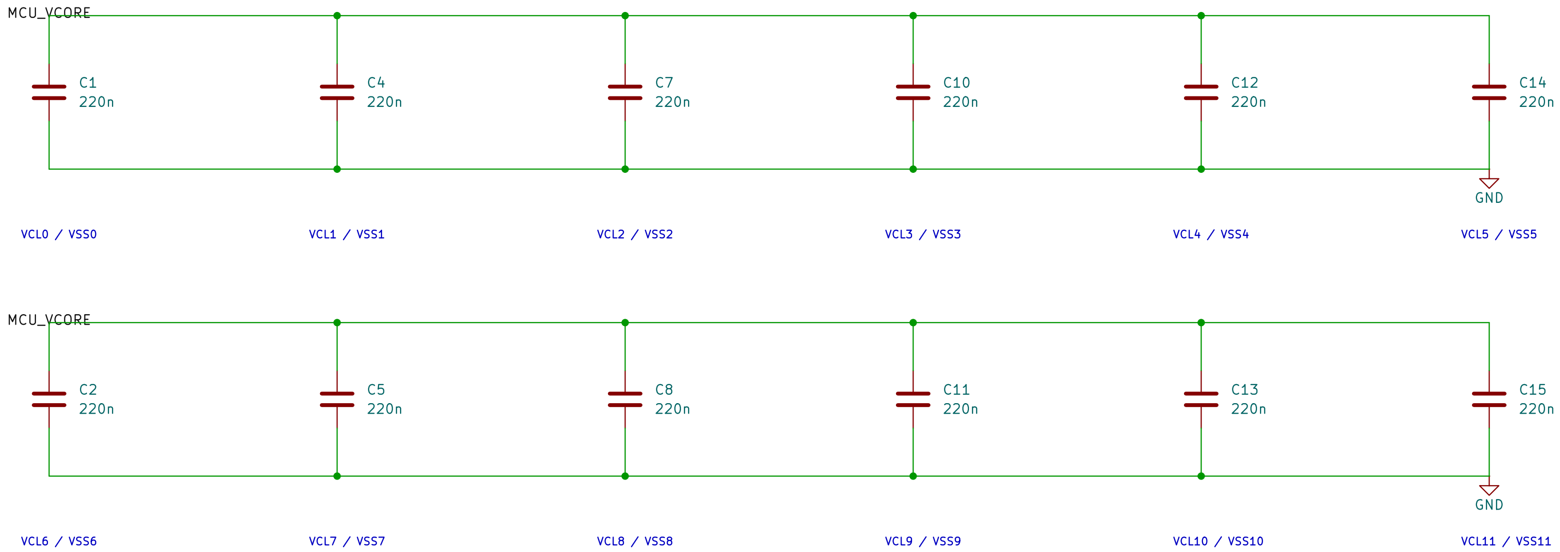
USB-003: USB-HS supply / U1.H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1+/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1+/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader
Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch
Title: RA8P1 IO allocation
Size: A3 | Date: 2026-09-05 | **Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6 | Id: 2/12

CORE DECOUPLING – 220 nF AT EACH VCL / VSS PAIR

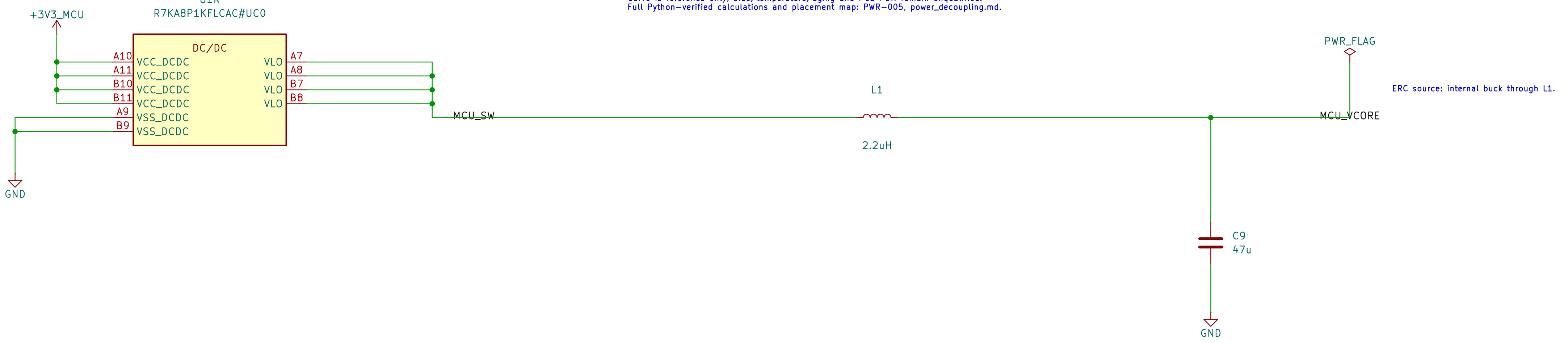


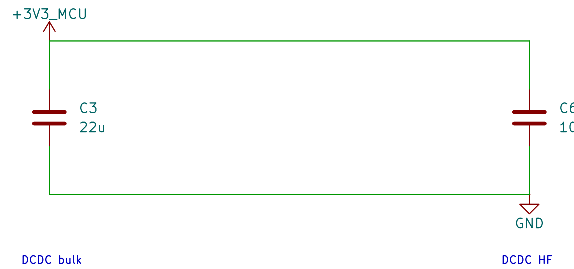


OFS2.DCDCEN = 1. All VLO balls share MCU_SW.
L1 feeds every VCL ball; never tie MCU_VCORE to 3.3 V.
C9 returns to VSS_DCDC. VCC_DCDC is tied to VCC.
HUM Rev.1.30, Table 69.2 / Fig.69.1, pp.4042–4044.

PWR–005 | LOCAL VCL BYPASS – C1,C2,C4,C5,C7,C8,C10–C15
220nF per VCL/VSS pair: Renesas QDG Rev1.10 section1.2 (p.8).
TDK C1608X7R1H224K080AB: 50V X7R +/–10%, twelve local capacitors.
C_initial = 220*(1 +/– .10) = 198..242nF each.
Bank = 12*220nF = 2.64uF nominal; 2.376..2.904uF initial.
TDK nominal curve: C(1.0V) = 220+(1/1.25)*(223.065–220) = 222.452nF.
Curve is reference only; bias/temperature/aging and PCB PDN remain unqualified.
Full Python–verified calculations and placement map: PWR–005, power_decoupling.md.

INTERNAL BUCK REGULATOR





L1 DCR <=100mOhm. Effective capacitance and inductor current
rating require final MPN qualification. +3V3_MCU is supplied by U13
on Main 3V3 digital supply; upstream SYS_AON source remains open.

RA8P1 e-reader
Sheet: /RA8P1 internal core regulator/
File: mcu_core_power.kicad_sch

Title: RA8P1 internal core regulator

Size: A3 | Date: 2026-09-05 | **Rev: 0.3 DRAFT**
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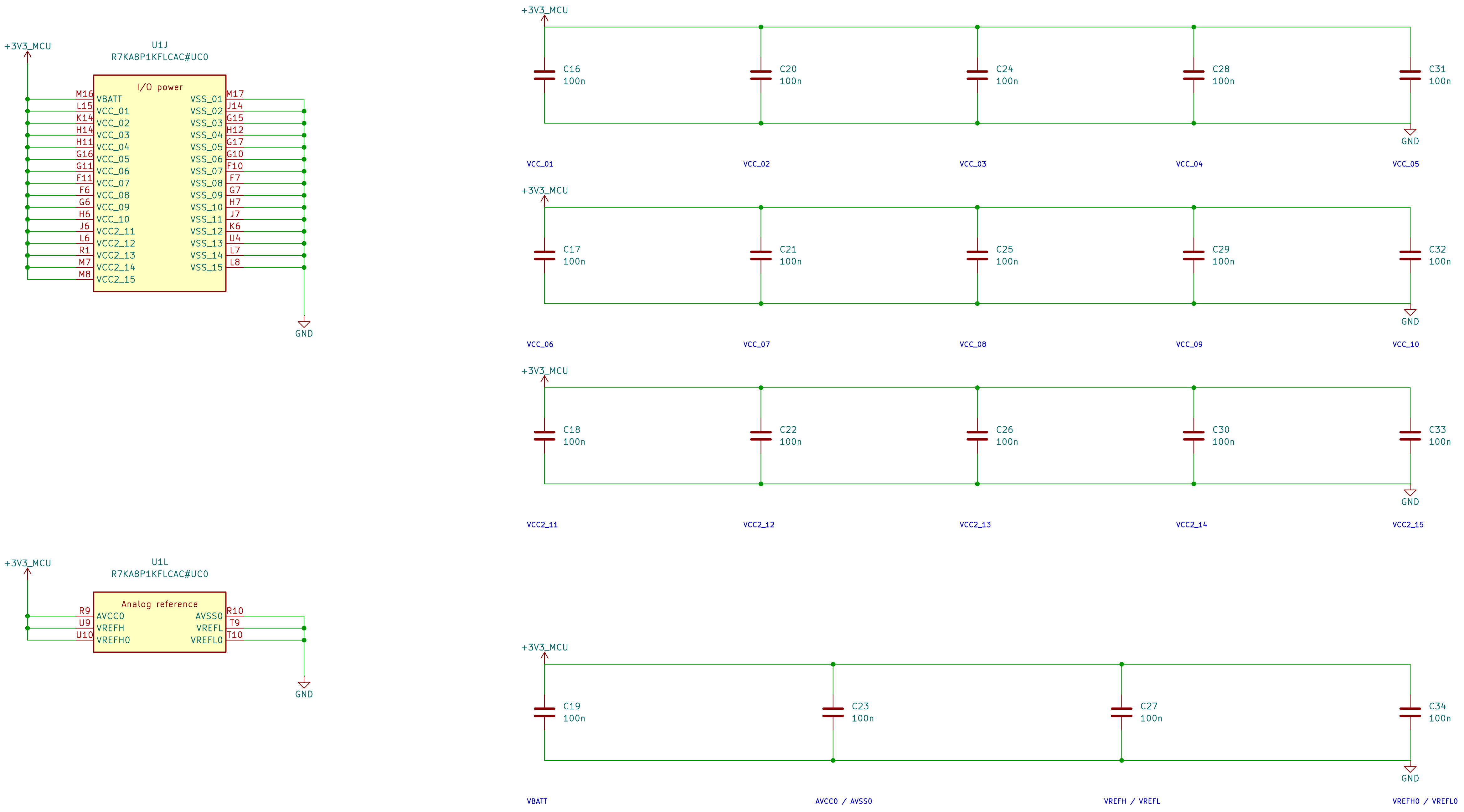
I/O DECOUPLING — 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIPI supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.

RA8P1 e-reader
Sheet: /RA8P1 IO and analog supplies/
File: mcu_supply_power.kicad_sch

Title: RA8P1 I/O and analog supplies

Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
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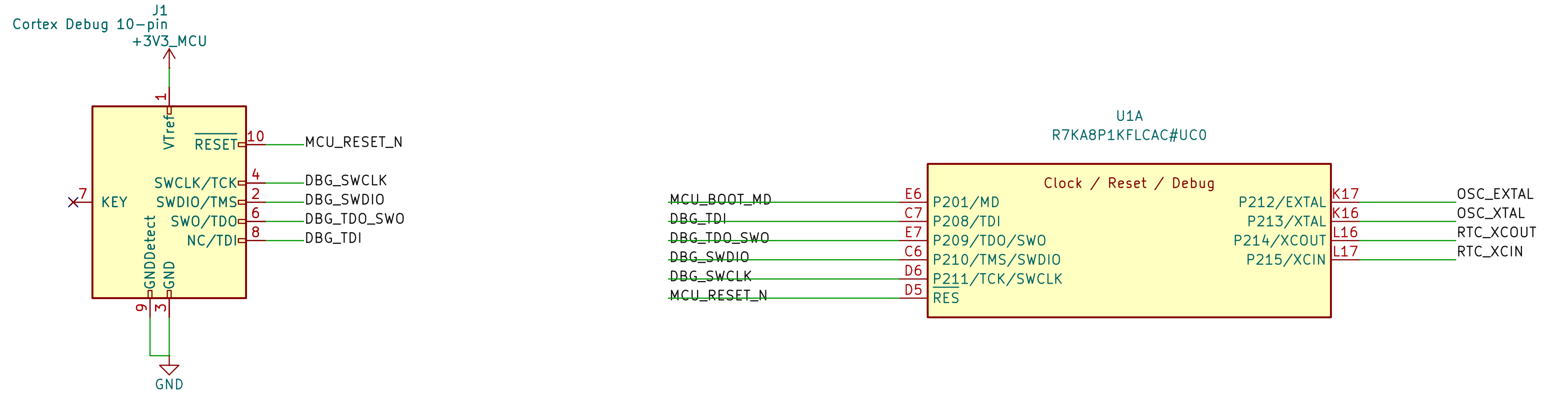
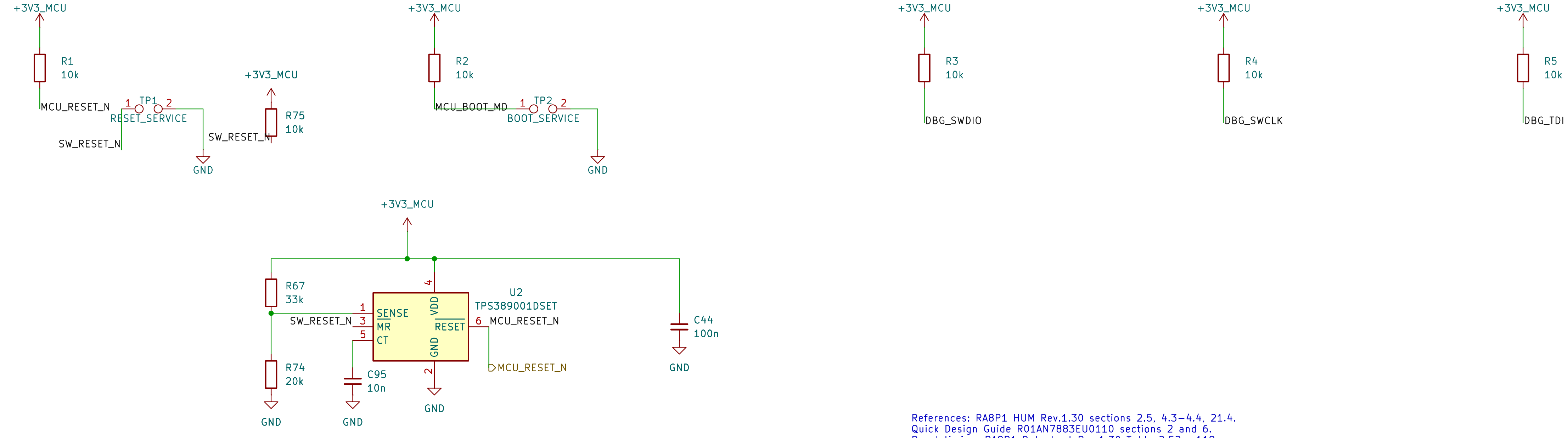


CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md



References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

RST-002: U2 TPS389001DSET; R67=33k, R74=20k, R75=10k, C95=10n.
Sense resistors: +/-0.05%, 10ppm/C over 100C, allocated +/-0.15% assembly/aging.
fL=.9995*.999*.9985=.99700274925; fH=1.0005*1.001*1.0015=1.00300275075.
Klo=1+33000*fL/(20000*fH); Khi=1+33000*fH/(20000*fL); E=150nA*33000*fH.
Vfall=1.15*.99*Klo-E .. 1.15*1.01*Khi+E = 3.000822727..3.094473285V.
Vrise=1.157*.99*Klo-E .. 1.157*1.01*Khi+E = 3.019118825..3.113278988V.
Main release margin=3.151819680-3.113278988=38.540692mV; no double hysteresis.
C95: 10n C0G +/-5%, 30ppm/C over 100C; allocate +/-10nA external CT leakage.
tcharge,min=10n*.95*.997*1.17/(1.35u+10n)=8.148276ms > 2.4ms RES minimum.
tcharge,max=10n*1.05*1.003*1.29/(.90u-10n)=15.264758ms; NOT total-delay max.
R75 MR pull-up: R=9801..10201ohm; allocate +/-50uA total adverse MR current.
MR high margin=.3*3.151819680-50uA*10201=.435495904V; sink <=.396190439mA.
Conditional DC/startup design, not fast-collapse or SDRAM-quiescence proof.
Full derivation, assumptions, sourcing and Python: design/reset_coordination_tps3890.md.

Sheet: /RA8P1 clocks, reset and debug/
File: mcu_clocks_debug.kicad_sch

Title: RA8P1 clocks, reset and debug

Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
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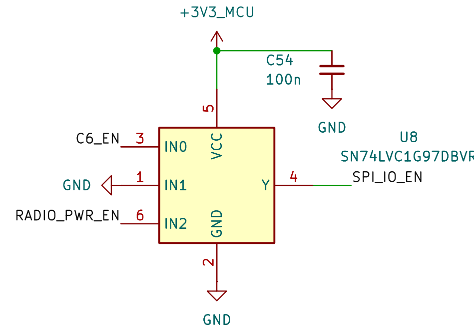
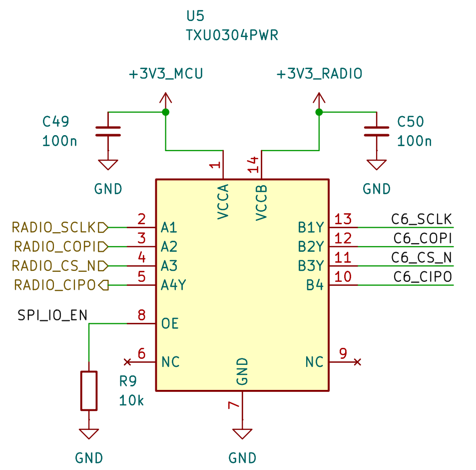
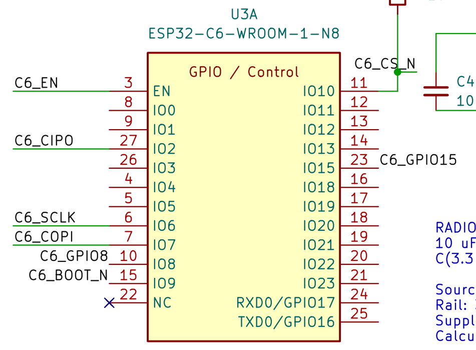
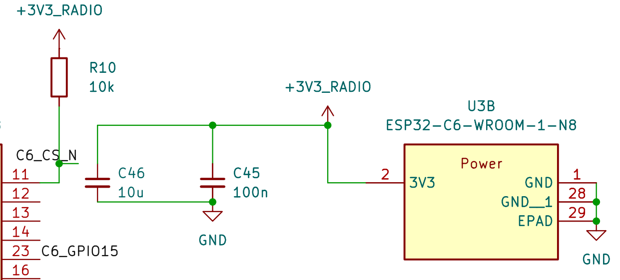


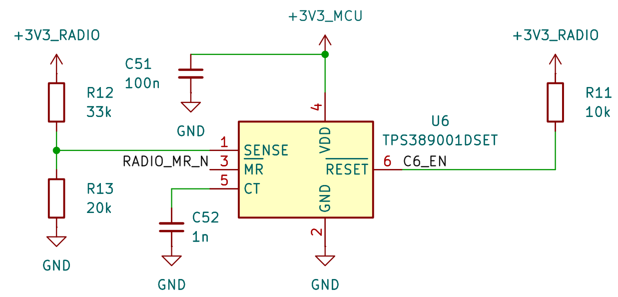
RADIO-016: SPI_IO_EN = C6_EN AND RADIO_PWR_EN
R9 bounds: 9.801k..10.201k (1%, 100ppm/K, 100K).
Ihigh <= 3.6/9801 + 12uA = 0.379309mA.
3V test-point margins: 2.4-1.92=0.48V high;
0.89-0.45=0.44V low. Full-rail sequencing OPEN.
12uA/node allocation includes U8; other loads unqualified.
Calculations: ../design/radio_interface.md#radio-016-spi-output-enable-arbitration

RADIO-012: C52 / U6 reset-release delay
C52 = 1nF +/-5% NP0, CT to GND (not VIN).
At 100 C excursion, +/-30 ppm/C: 947.15..1053.15 pF.
tnom = 1nF*1.23V/1.15uA + 25us = 1.094565 ms.
Allocate +/-10nA external CT leakage; start discharged.
tcharge_min = 947.15pF*1.17V/1.36uA = 814.828 us.
tcharge_max = 1053.15pF*1.29V/0.89uA = 1526.476 us.
Charge-only screen, NOT total reset min/max limits.
C6 requires 50us stable supply and 50us reset low.
Complete reset sequencing remains open.
Math: ../design/radio_interface.md#radio-012-supervisor-reset-release-delay



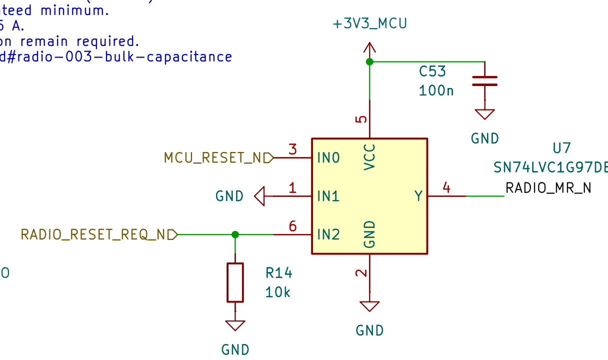




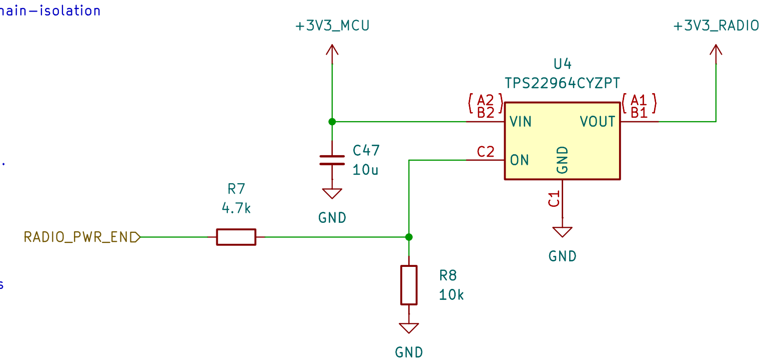


RADIO-003: C46 bulk bypass / TDK C3216X7R1V106K160AC
10 uF nominal, +/-10%, X7R; DC bias reduces capacitance.
C(3.3 V) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)
= 9.64418 uF; C(3.6 V) = 9.53503 uF (nominal).
Source curve interpolation, NOT a guaranteed minimum.
Rail: 3.0..3.6 V; source capability >= 0.5 A.
Supply and transmit transient qualification remain required.
Calculations: ../design/radio_interface.md#radio-003-bulk-capacitance

RADIO-014 / RST-002: R12=33k, R13=20k switched-radio SENSE divider.
0.05%, 10ppm/C; allocate 100C, +/-0.15% installed drift, +/-150nA total.
fL=.9995*.999*.9985; fH=1.0005*1.001*1.0015.
Klo=1+33000*fL/(20000*fH); Khi=1+33000*fH/(20000*fL).
E=150nA*33000*fH; Vtrip=Vthreshold*(1+Rt/Rb)+Is*Rt.
Vfall=1.15*.99*Klo-E .. 1.15*1.01*Khi+E = 3.000822727..3.094473285V.
Vrise=1.157*.99*Klo-E .. 1.157*1.01*Khi+E = 3.019118825..3.113278988V.
Use direct VITP limits; do not add hysteresis a second time.
RADIO-019 U4 TPS22964C: 3.151819680-.020-3.113278988
=18.540692mV conditional release margin; <=20mV path at 505mA required.
30mOhm installed switch allocation needs full-range qualification.
No radio DC/startup or fast-brownout sign-off from this model.
C52=1n charge interval .814828..1.526476ms; not total IC delay maximum.
Full assumptions, sourcing and Python: design/reset_coordination_tps3890.md.



RADIO-009: U5 isolates host and switched-radio signal domains.
C49/C50: 100 nF local bypass; TI TXU0304 section 12.1.
C50 adds 0.1 uF: fitted external radio capacitance = 10.2 uF nominal.
5 MHz budget: 100 - 11 - 11 = 78 ns remaining (5 pF test load).
SPI/OE and reset qualification remain open; timing is not closed.
Math: ../design/radio_interface.md#radio-009-spi-power-domain-isolation



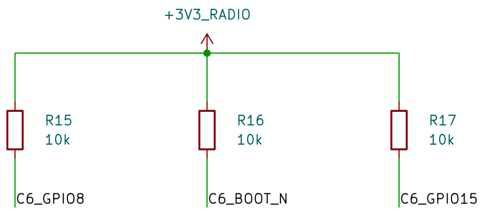
PWR-006: upstream U13 TPS63806YFFR, forced PWM.
Vnom=.5*(1+5600/(1000+10))=3.272277228V.
Static resistor/reference/leakage screen: 3.226819680..3.318012496V.
With one +/-75mV dynamics allocation: 3.151819680..3.393012496V.
RST-002 MCU rising maximum 3.113278988V leaves 38.540692mV.
RADIO-019 U4 TPS22964C: require <=20mV complete path at 505mA.
Radio release margin =38.540692-20=18.540692mV, conditional.
Continuous-rail RON, inrush, collapse and discharge need qualification.
Full regulator calculations: design/main_regulator_tps63806.md.
Reset calculations: design/reset_coordination_tps3890.md.

RADIO-010: R9/R10 idle-state bias, 10k 1%
Rmin/max = 10k*(1 +/-1%)*(1 +/-1%) = 9801/10201 ohm.
12 uA adverse leakage is an ALLOCATION, not measured.
R9: Vlow = 12u*10201 = 0.122412 V.
TI VT- min = 0.17 V at the 1.1 V equal-rail test point.
R10 @3.3 V: VCS >=3.177588 V > VIH 2.475 V (25 C).
R9 enabled current = 3.3/10k = 0.330 mA.
R10 draws similar current only while CS is low.
OE arbitration, all-state leakage and timing remain open.
Math: ../design/radio_interface.md#radio-010-idle-state-bias

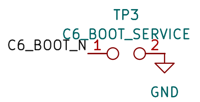
RADIO-019: U4 TPS22964CYZPT radio power gating
RST-002: 3.151819680 - 0.020 - 3.113278988
= 18.540692 mV conditional reset-release margin.
505 mA: switch <=30 mOhm + rest <=9.6 mOhm
required for <=20 mV path; qualification pending.
Internal slew: 715 us TYP; no CT capacitor.
QOD: 273 ohm TYP / 325 max; external C = 10.2uF.
Module C and leakage unknown; no guaranteed off-time.
RCB only disabled, either rail >1 V; decay unqualified.
Math: ../design/radio_interface.md (RADIO-019).

RADIO-019: R7=4.7k / R8=10k, active-high ON, default OFF
1% initial x 100ppm/C x 100C; adverse Ileak=+/-12uA.
VON=(Vhost/R7+Ileak)/(1/R7+1/R8); use resistor corners.
Valid host VOL<=0.5V / VOH>=2.5V; pull-up disabled.
Vlow<=0.383594V <0.4V; Vhigh>=1.639635V >1.3V.
Host source current <=0.261871mA in this envelope.
Hi-Z: 12uA*10201=0.122412V; leakage allocation required.
No logic guarantee once host leaves its valid voltage range.
Source collapse, startup and shutdown qualification remain open.
Math: ../design/radio_interface.md (RADIO-019).

RADIO-013: R11 / C6_EN reset pull-up
10k 1%, 100ppm/C; deltaT=100 C allocation.
Rmin/max=10k*(1 +/-1%)*(1 +/-1%)=9801/10201 ohm.
Allocate +/-12uA total adverse node leakage.
Isink <=3.6/9801+12uA=0.379310mA <0.4mA.
U6 VOL<=0.25V at 0.4mA, VDD>=1.5V.
At C6 table point 3.3V, 25 C:
VEN>=3.3-12uA*10201=3.177588V >2.475V VIH.
VIL=0.825V; low margin=0.825-0.25=0.575V.
Pmax=3.6^2/9801=1.322314mW.
MR arbitration, power states and timing remain open.
Math: ../design/radio_interface.md (RADIO-013)



RADIO-005: C47 / U4 VIN local bypass
10 uF nominal, +/-10%, 35 V X7R; same exact TDK part as C46.
C(3.3 V) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)
= 9.64418 uF nominal; NOT a guaranteed minimum.
Droop = integral(Iload-Isource)dt / Ceff + ESR/ESL effects.
TI section 11: 1 uF usually sufficient; slow sources may need more.
C47 is upstream: exclude it from C45+C46 output-discharge math.
Regulator/transient qualification required; no source flag added.
Calculations: ../design/radio_interface.md#radio-005-input-bypass



RADIO-015: U7 Schmitt-input reset arbitration
IN1=GND: Y = MCU_RESET_N AND RADIO_RESET_REQ_N.
Either low asserts U6 MR; C53 = 100nF MCU-rail bypass.
R14: Rmin/max=10k*(1 +/-1%)*(1 +/-1%)=9801/10201 ohm.
12uA adverse-current ALLOCATION: Vlow <= 0.122412 V.
At U7 3V test point: VT-_min-Vlow = 0.717588 V.
At 3.6V: I_R14 <= 0.367309 mA; P <= 1.322314 mW.
If |I_U7out|<=100uA: MR low/high margins >=0.65/0.8V
on shared 3.0-3.6V rail. MR load bound NOT yet verified.
Reset hierarchy connected; full-rail thresholds/sequencing open.
Math: ../design/radio_interface.md#radio-015-reset-request-arbitration

RADIO-019: internal slew / conditional discharge screens
TI tR(10..90%)=715us TYP at 3.3V, 25C, 10 ohm load,
CIN=1uF/COUT=0.1uF; not this board or worst-case timing.
External C45+C46+C50=10.2uF nominal; module C unknown.
Icap=10.2uF*(0.8*3.3V)/715us=37.661538mA (linear TYP).
t90..10=R*C*ln(9): 6.118392ms at 273 ohm TYP;
7.283799ms using 325 ohm table max at ON=0, IOUT=2mA.
Ideal RC only; C corners, leakage, VIN decay need qualification.
Keep radio OFF during host cold/wake inrush; wait for rail settle.
Verify U6/C6 reset interval independently; no fixed off-time.
Math: ../design/radio_interface.md (RADIO-019).

RADIO-017: R15/R16/R17 boot/debug strap pull-ups to switched radio rail.
Flash boot: GPIO9=1. Download: GPIO8=1, GPIO9=0; service pads TP3 (RADIO-018).
GPIO15=1 selects USB JTAG with applicable eFuses; retain straps >=3ms after EN rises.
Rmin/max=10k*(1+/-1%)*(1+/-100ppm/K*100K)=9801/10201 ohm.
At 3.3V: Vhigh>=3.3-12uA*10201=3.177588V; VIH=2.475V; margin=0.702588V.
External low-state current<=3.6/9801=0.367309mA each; P<=1.322314mW each.
12uA total/node is an allocation. Internal pull-up sink current and full-state timing OPEN.
Math: ../design/radio_interface.md#radio-017-boot-and-debug-strap-bias

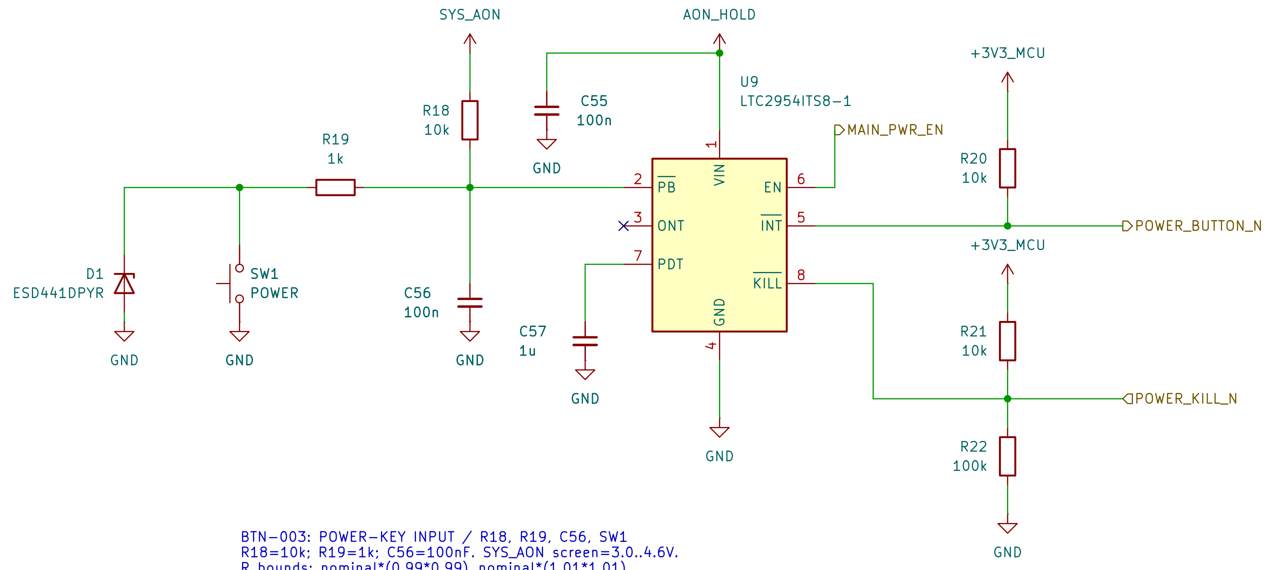
RADIO-018: TP3 internal C6 BOOT service pads; no fitted switch.
Fixture shorts pad 1 (GPIO9) to pad 2 (GND); GPIO8 must be high.
Hold GPIO9 low before C6_EN rises and >=3ms after C6_EN rises.
R16 external current <=3.6/9801=0.367309mA; add internal pull-up.
Fixture ALLOCATION: total sink <=1mA and path <=1ohm => Vlow<=1mV.
At 3.3V: low margin=0.25*3.3-0.001=0.824V (screen only).
Host-independent radio power/reset arbitration remains to be connected.
Math: ../design/radio_interface.md#radio-018-internal-boot-service-pads

Incomplete electrical design - not for manufacture

Sheet: /ESP32-C6 radio/
File: radio_esp32.kicad_sch

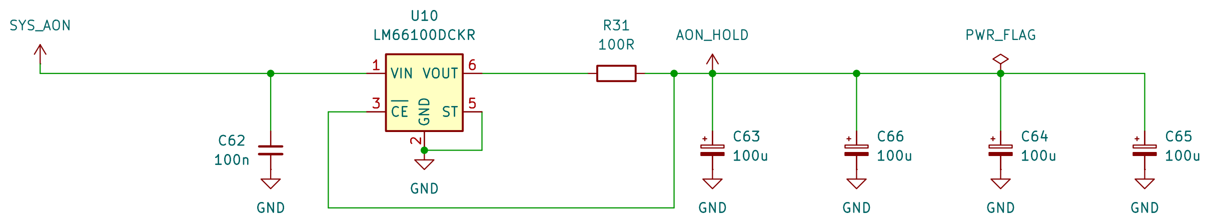
Title: ESP32-C6 radio

Size: A3 | Date: 2026-09-05 | Rev: DRAFT
KiCad E.D.A. 10.0.6 | Id: 6/12



INCOMPLETE POWER CONTROL – NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main–rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 EN.
SYS_AON source and USB permission–latch clear remain to be built.
Current converter: U13 TPS63806; design/main_regulator_tps63806.md PWR–006.
MCU reset: design/reset_coordination_tps3890.md RST–002.
Radio release margin and system shutdown coordination remain open (#846).

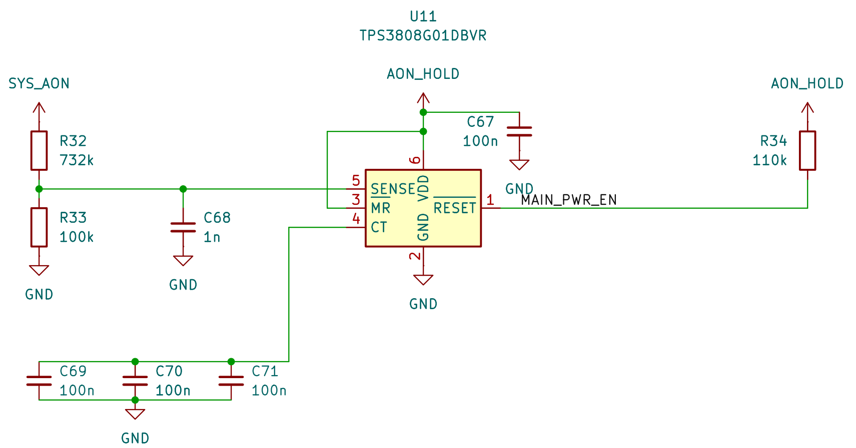
ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series–fed source only;
raw SYS_AON source remains unimplemented.



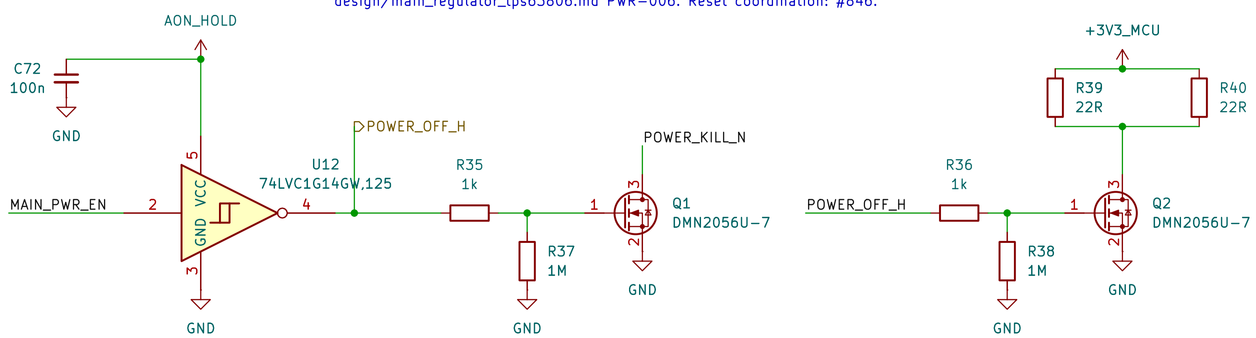
BTN–003: POWER–KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6–0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(10201+1020.1+0.5)–12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN–003/005/010.

BTN–004: HARDWARE FORCED–OFF TIMING / C57
C57=1uF; TDK C2012X7R1E105K125AB, 25V X7R +/–10%.
C[uF]=1.56e–4*(t_added[ms]–1).
t_force,nom=64ms + 1ms + 1/1.56e–4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF;
1*1.10*1.15=1.265uF –> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn–on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN–004/005.

SYS–007 / PWR–004: HELD SUPPLY / U10, R31, C62–C66
C63–C66: 4 x 100uF/10V; Cmin=400uF*0.9^3=291.6uF.
VIN=raw SYS; VOUT–>R31–>reservoir; CE senses reservoir.
Icontrol<=1.125mA incl 500uA bank leakage allocation.
Pre–trip reverse: 80mV/98.01ohm<0.817mA. Itotal=1.942mA.
Vinitial=3.263705831–max(0.250,0.001125*102.24)
=3.013705831V (threshold OR resistive–drop limit).
t_hold=[291.6uF*(3.013705831–2.7)V–1uC]/1.942mA
=46.589403ms >38.997458ms shutdown at 1mF.
Key–return–aware hold margin=7.591945ms at 1mF.
1uC covers initial undershoot + reverse/gate transitions.
Qualify VIN=0..4.6V, slow/floating–source collapse and load.
R31 short: 4.6^2/98.01<0.216W; 1W nominal rating.
Control only: no MCU/audio/display loads on AON_HOLD.
Separate unadopted eFuse extra load is not included.
Sources/Python: ../design/system_power_design.md SYS–007;
../design/main_regulator_ltc3119.md PWR–004.



SYS–007 / PWR–006: SOURCE SUPERVISION / U11, R32–R34, C67–C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/–2%, Isense +/–25nA; each R +/–0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69–C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k: ENhigh>=2.7–3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3–0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer; held Q3 clamps TPS63806 EN.
Held U12 also drives KILL/discharge; USB–clear still needed.
Current EN network and rail calculations: PWR–006.
Sources/Python: design/system_power_design.md SYS–007;
design/main_regulator_tps63806.md PWR–006. Reset coordination: #846.

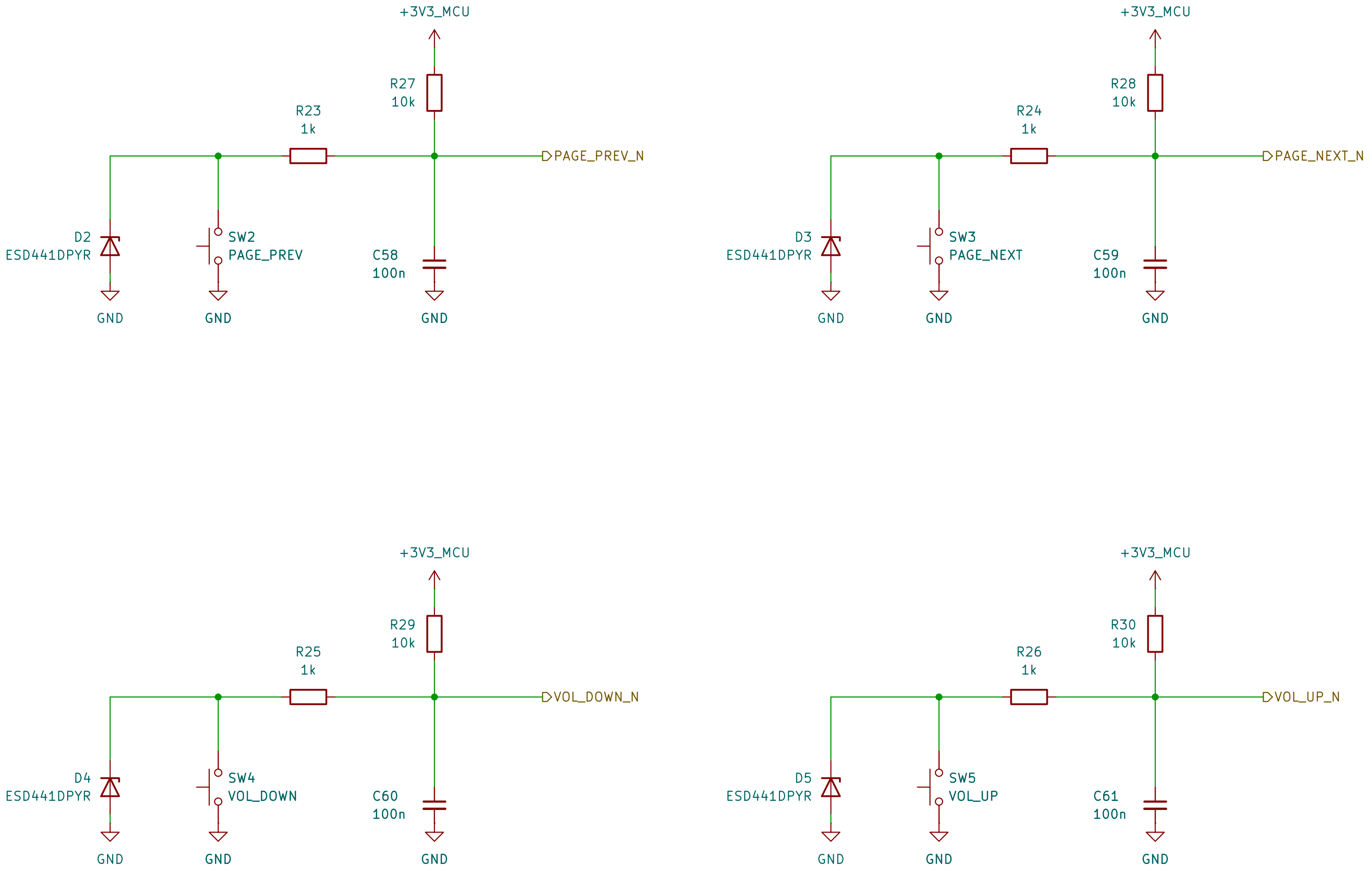


SYS–007 / PWR–004: SHUTDOWN / U12, Q1–Q3, R35–R40, R70–R71
POWER_OFF_H=NOT(MAIN_PWR_EN); held–domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% + 100ppm/C, 100C.
4–gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB–clear; static<=22.773595uA <100uA.
VGSmin=(2.6–1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage is a qualification allocation.
RUN/KILL/discharge/USB–clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Key–filter return allocation Ir=1.5mA; R=Rmax.
Total=10ms+R*C*ln((3.6–Ir*R)/(0.3–Ir*R)).
At 850uF: 34.647839ms; at 1mF: 38.997458ms <46.589403ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W; 1W at 70C.
Qualify 0.2ohm FET, 10ms TOTAL response, derating and backfeed.
Gate/transient charge shares the existing 1uC reserve.
VBATT firmware/option invariant remains unimplemented; #846 unresolved.
Sources/Python: ../design/system_power_design.md SYS–007;
../design/main_regulator_ltc3119.md PWR–004.

Sheet: /Power button and source control/
File: power_button.kicad_sch

Title: RA8P1 E–reader – power button and source control

Size: A3	Date: 2026–09–07	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 7/12



CMS−009 − FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU: 3.0..3.6V screening envelope.
Resistors: +/−1%, plus +/−100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=980.1..1020.1 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3−2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)−2.2uA >265.14uA.
Open contact>2.97745V: switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS−009.

CMS−009 − FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature−only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=−tau_assert*ln[(0.2−1/11)/(1−1/11)]=0.193ms.
t_release=−10k*100n*ln[(1−0.8)/(1−1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot−held ordinary actions until release; no auto−repeat
from a stuck key. Disable MCU internal pulls.
In hard−off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced−off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

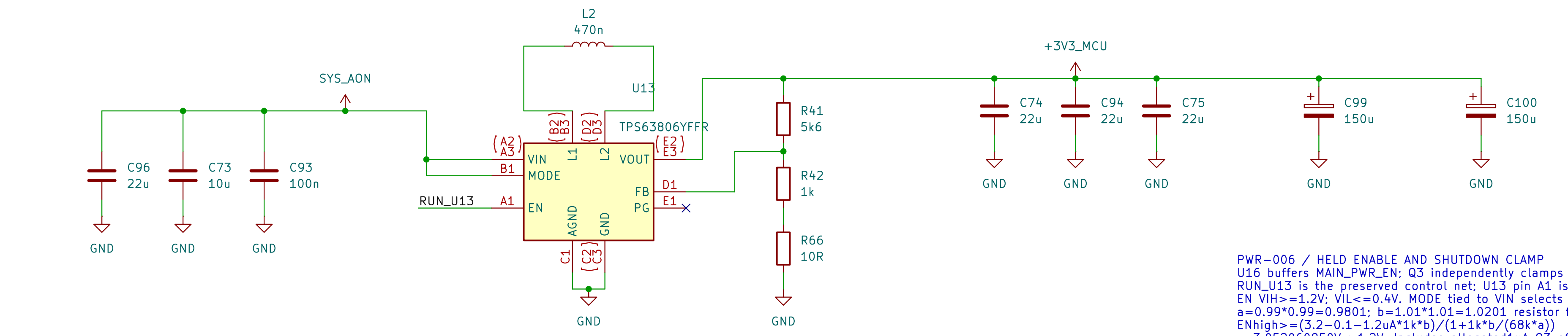
Full corner calculations, Python and wake−register contract:
../design/camera_storage_interfaces.md CMS−009.

Sheet: /Page and volume buttons/
File: user_controls.kicad_sch

Title: RA8P1 E−reader − page and volume buttons

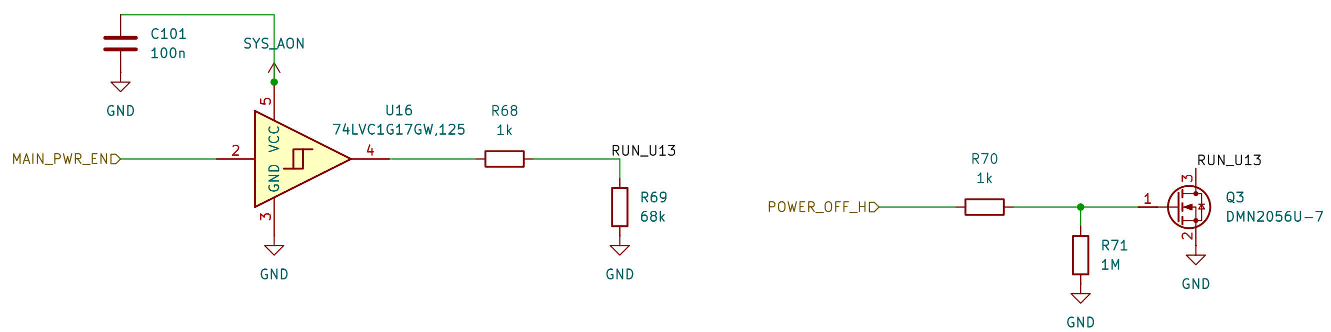
Size: A3 | Date: 2026−09−07 | **Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6 | Id: 8/12

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



PWR-006 / HELD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWR_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
a=0.99*0.99=0.9801; b=1.01*1.01=1.0201 resistor factors.
ENhigh>=(3.2-0.1-1.2uA*1k*b)/(1+1k*b/(68k*a))
=3.052060850V >1.2V. Includes allocated1uA Q3 off leakage.
U16 high load<=4.6/(68k*a)+1.2uA=70.220568uA <100uA.
ENlow<=0.1+0.2uA*1k*b=0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 Ioff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*a)+1uA)/(1/(1k*a)+1/0.2ohm)
=0.938689mV <0.4V; ignores helpful R69.
TI0.2uA EN leakage applies only under its powered test conditions.
Retain1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Ron are acceptance allocations.

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*4.6/(1k*a)=18.773595mA (not continuous rating).
Isteady<=4*(4.6/(1M*a)+1uA)=22.773595uA.
VGS>=(2.7-0.1-1uA*1k*b)/(1+1k*b/(1M*a))=2.596278V.
Held control spare=1125-500-500-4-22.773595=98.226405uA.
500uA terms: reservoir leakage and non-rail inverter-input allocation.
Retain1uC transition charge and10ms complete response allocation.
R34=110k: POR sink=(1.3-0.2)/(110k*a)+3.8uA=14.003041uA.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Full assumptions, primary sources and Python verification: click this note.



PWR-006 / FEEDBACK AND LOCAL PASSIVES
Vnom=0.500*(1+5600/(1000+10))=3.272277228 V.
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed bounds: initial*TCR(100C)*aging*assembly, then ohms.
R41=5600*(1+/-0.0002)*(1+/-0.0005)*(1+/-0.001)
*(1+/-0.0005)+/-0.02 = 5587.669237..5612.349243 ohm.
R42=1000*(1+/-0.0001)*(1+/-0.0002)*(1+/-0.001)
*(1+/-0.0005)+/-0.01 = 998.190970..1001.810970 ohm.
R66=10*(1+/-0.01)*(1+/-0.02)*(1+/-0.01)+/-0.05
=9.554980..10.455020 ohm; one combined aging/assembly 1%.
Vlo=0.495*(1+R41min/(R42max+R66max))-100nA*R41max.
Vhi=0.505*(1+R41max/(R42min+R66min))+100nA*R41max.
Vstatic=3.226819680..3.318012496 V (FPWM only).
One combined +/-75mV regulation/ripple/transient/routing allowance:
Vcomplete=3.151819680..3.393012496 V; qualification required.
SDRAM static VOH margin=2.4-0.7*3.393012496=24.891253mV.
This is not measured SI/noise headroom or reset qualification.

L2: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
=>0.3708..0.5698uH versus TI0.37..0.57uH. Qualify hot/current.
C74/C75/C94: 3*22uF*0.8*0.85*0.6=26.928uF > TI21uF.
C96: 22uF*0.8*0.85*0.6=8.976uF > TI4uF input minimum.
0.6 is allocated combined bias/aging retention, NOT guaranteed.
Samsung typical output-bias retention at3.393012496V=90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LOAD, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold-start screen, not measured load.
Pout=3.393012496*1.8=6.107422493W.
At allocated75% efficiency: Pin=8.143229991W; Iin@3.2V=2.544759372A.
Loss=2.035807498W: NOT an acceptable continuous thermal result.
At90% efficiency: loss=0.678602499W.
TJ screen=70C+78.8C/W*0.678602499W=123.474C.
TI board theta-JA is not the final PCB thermal resistance.
2.25A screen: Pin=3.393012496*2.25/0.75=10.179037489W.
Converter limit, source limit, startup and thermal envelope need validation.

Native storage inventory (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on adds10.2uF =>489.81uF.
C99/C100 remain150uF each. C94 adds22uF to the output.
Old bootstrap/private VCC/VC capacitors were not main-rail storage.
Whole-main ceiling remains1mF for shutdown; no LTC stability transfer.
Cout lower acceptance and distributed PDN/startup are separate checks.

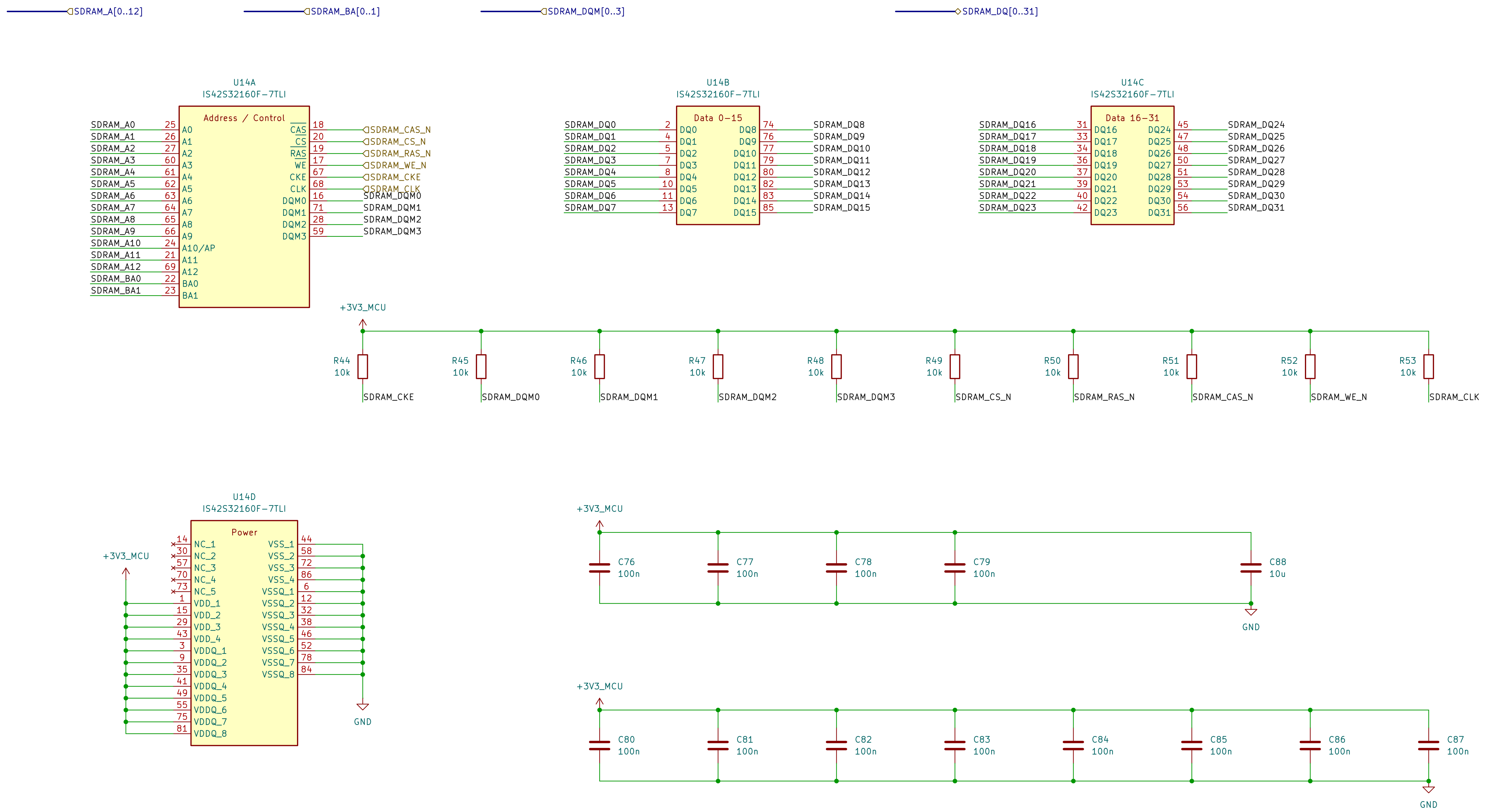
Preserved discharge screen: Rmax11.4211ohm; key return<=1.5mA.
Vinf=1.5mA*11.4211ohm=0.01713165V.
t=1mF*11.4211ohm*ln((3.6-Vinf)/(0.3-Vinf))+10ms
=38.997458ms to0.3V, below held46.589403ms screen.
Response10ms includes command/clamp/discharge delays.
Held budget retains1uC transition charge and98.226405uA spare.
SYS_AON source protection, loaded startup and fast faults remain open.
RST-002 / RADIO-019: release margin remains conditional on rail/path bounds.
Do not power up until reset coordination and source protection are complete.
Full assumptions, sources and executable Python: click this note.

Sheet: /Main 3V3 digital supply/
File: main_3v3_supply.kicad_sch

Title: RA8P1 E-reader – main digital 3.3 V supply

Size: A3	Date: 2026-09-08	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 9/12

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset-default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.

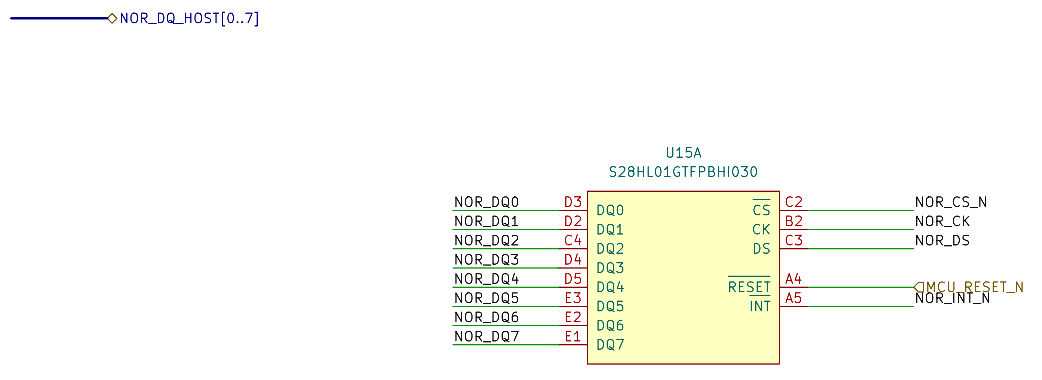


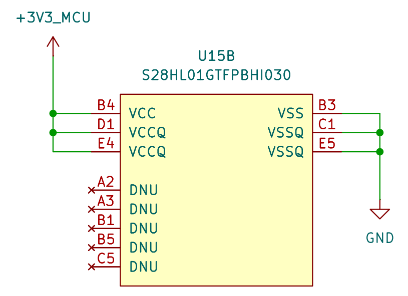
CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

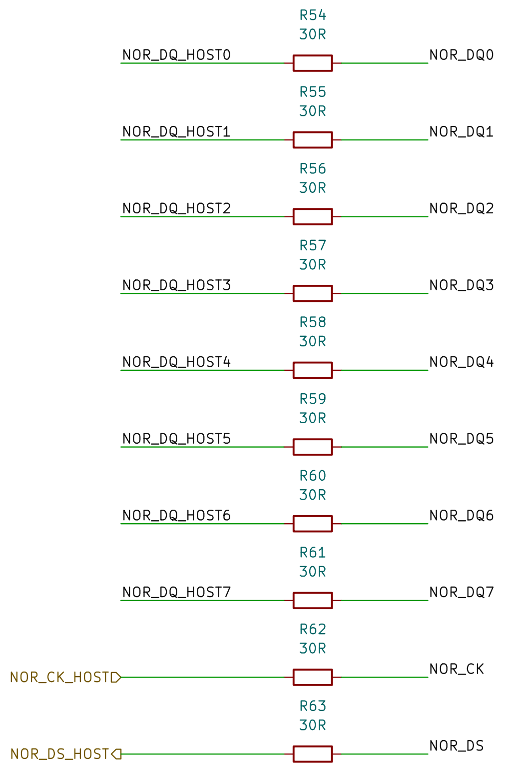
CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V; +/-1%, 100ppm/C, |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

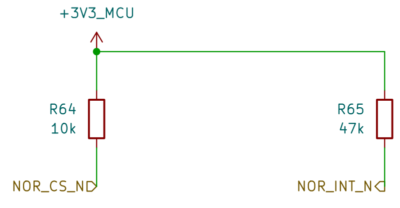
CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail;
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

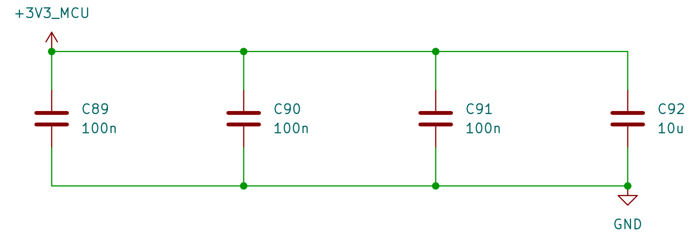
Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch
Title: RA8P1 E-reader - 64 MiB SDRAM
Size: A3 | Date: 2026-09-08 | **Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6 | Id: 10/12











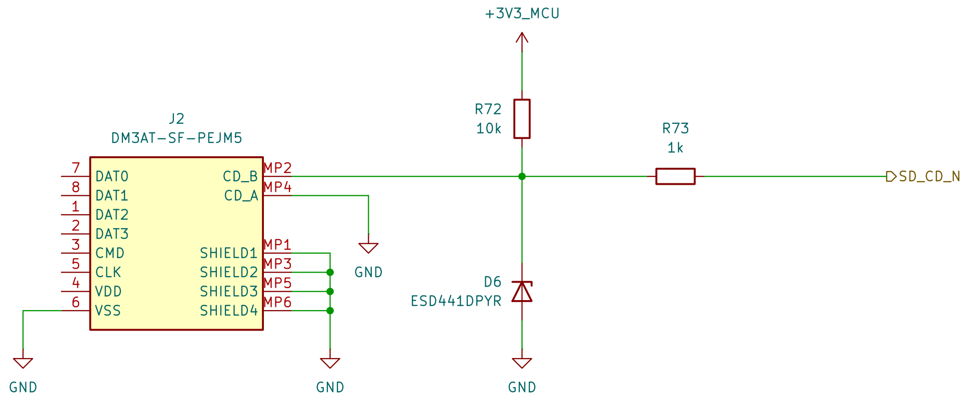
CMS−011 | NOR series and idle states
R54−R63: 10 x 30R, 0.1%, 25 ppm/C.
100 C departure from reference temperature:
Rmin = 30 x 0.999 x 0.9975 = 29.895075 ohm
Rmax = 30 x 1.001 x 1.0025 = 30.105075 ohm
Within JESD251 30R +/−1%: 29.7..30.3 ohm.
External parts remain required; SI check OPEN.

R64 CS pull: 10k; R65 INT pull: 47k.
1%, 100 ppm/C, 100 C: factors 0.9801..1.0201.
At 4uA adverse leakage, rail min 3.242044 V:
CS Vhigh >= 3.242044 − 4u x 10201 = 3.201240 V
INT Vhigh >= 3.242044 − 4u x 47944.7 = 3.050265 V
INT Ilow <= 3.6/46064.7 + 2u = 80.150949 uA
< 100uA flash VOL test; 47k departs from 5−10k.
RC timing and board leakage qualification OPEN.

Linked CMS−011B/C/F: sources and Python checks.
MCU hierarchy and common reset connected.
DS is read−only: P801 input, WRMSKMD=0.
CMS−011D defines reset and protocol recovery.

CMS−011 | NOR supply and bypass
U15B VCC and VCCQ share +3V3_MCU (2.7 to 3.6 V).
C89−C91: 3 x 100 nF, one per supply ball.
C92: 10 uF local bulk.
Cnom = 3 x 0.1 + 10 = 10.3 uF.
Initial +/−10%: Cmin/max = 10.3 x (0.9 / 1.1)
= 9.27 / 11.33 uF; not an effective−C guarantee.
Included in the PWR−003 1 mF whole−rail limit.
DC bias, temperature, aging and PDN qualification OPEN.
Linked CMS−011B/F: sources and Python−verified arithmetic.

Sheet: /128 MiB Octal NOR/		
File: nor_flash.kicad_sch		
Title: RA8P1 E−reader − 128 MiB Octal NOR		
Size: A3	Date: 2026−09−08	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.6		Id: 11/12



CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k: 9801..10201 ohm; R73=1k: 980.1..1020.1 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; I_leak=2.1uA; contact Rc=0.14 ohm (qualification assumption).
Vhigh=VCC-I_leak*(R72+R73) >=2.976435V.
Vlow=VCC*Rc/(R72+Rc)+I_leak*(R73+R72||Rc) <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hirose 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off; >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; card power/data bus incomplete.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

Incomplete electrical design - not for manufacture

Sheet: /microSD removable storage/
File: microsd.kicad_sch

Title: RA8P1 e-reader - microSD removable storage

Size: A3	Date: 2026-09-08	**Rev: 0.3 DRAFT**
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